THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE CHAPTER TEST SET B

Chapter 4: **Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An atom of chlorine has atomic number 17 and mass number 35. The number of neutrons in it is: 1

(a) 17 **(b)** 35 **(c)** 19 **(d)** 18

2. An atom of carbon has atomic number 6 and mass number 12. The number of neutrons in it is: 1

(a) 6 **(b)** 7 **(c)** 5 **(d)** 12

3. The electronic configuration (K, L, M, N shells) of the potassium atom (Z = 19) is: 1

(a) 2, 8, 9, 0 **(b)** 2, 9, 8 **(c)** 2, 8, 8, 1 **(d)** 2, 8, 8, 2

4. The valency of aluminium (Z = 13) is: 1

(a) 2 **(b)** 3 **(c)** 4 **(d)** 0

5. The maximum number of electrons that the shell with n = 2 can hold is: 1

(a) 8 **(b)** 10 **(c)** 16 **(d)** 4

6. Cathode rays were shown by J. J. Thomson to be: 1

(a) protons **(b)** streams of positive ions

(c) streams of negatively charged particles **(d)** neutral waves

7. The nucleus is about _____ times smaller than the atom. 1

(a) 10^{10} **(b)** 10

(c) 10^{5} (one lakh) **(d)** 10^{2}

8. Which of the following particles has a negligible mass compared to others? 1

(a) neutron **(b)** electron **(c)** α-particle **(d)** proton

9. Which pair are isotopes? 1

(a) $^{12}_{6}\mathrm{C}$ and $^{14}_{7}\mathrm{N}$ **(b)** $^{1}_{1}\mathrm{H}$ and $^{4}_{2}\mathrm{He}$

(c) $^{40}_{18}\mathrm{Ar}$ and $^{40}_{20}\mathrm{Ca}$ **(d)** $^{35}_{17}\mathrm{Cl}$ and $^{37}_{17}\mathrm{Cl}$

10. Isotopes of an element have similar chemical properties because they have the same: 1

(a) number of neutrons

(b) physical properties

(c) mass number

(d) number of electrons and electronic configuration

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. State two conclusions of Rutherford's gold foil experiment. 2

12. State one limitation of Rutherford's atomic model. 2

13. State two postulates of Bohr's model. 2

14. Why was the neutron needed to explain atomic masses? Give an example. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. How many atoms of diameter 10^{-10} m must be stacked to make a sheet 0.1 mm thick? 3

16. Explain how Bohr's model removed the difficulty of Rutherford's model. 3

17. Why are the shells called K, L, M, N and not A, B, C, D? 3

OR

How did Chadwick's discovery explain why a helium atom is four times heavier than hydrogen? Why are neutrons important in heavy nuclei? 3

18. Why are the symbols of elements written using Latin names? Give two examples with their norms. 3

OR

An atom has 20 protons and its mass number is 40. Find the number of neutrons and electrons, its configuration and valency. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A hydrogen atom has one proton and a helium atom has two protons, yet helium's mass is four times that of hydrogen. 5

(i) What puzzle does this pose?
(ii) Who resolved it and how?
(iii) What are the nucleons?
(iv) Why do heavier atoms contain many more neutrons than protons?

OR

(i) State the rules of Bohr–Bury for distribution of electrons. 5
(ii) Write the electronic configuration of Mg, Cl, S, K and Ca.
(iii) Give their valencies.

20. **Case study:** An atom X has atomic number 17 and mass number 35. 5

(i) Find protons, neutrons and electrons.
(ii) Write its electronic configuration.
(iii) Find its valency and name the element.
(iv) Predict whether it forms a cation or anion.

OR

(i) Define valency and valence electrons. 5
(ii) Explain with Na, O and C how valency is determined.
(iii) Why are noble gases unreactive?

END OF QUESTION PAPER — **Total: 40 Marks**